

AiM MXP SmartyCam → CANable 2.0 → Pixel 10

Integration Validation

2003 BMW M3 (E46) — Live Bus Capture Report (v3.0 Spec)

Pitwall Engineering

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Abstract

End-to-end validation of the AiM MXP SmartyCam ingest chain on a 2003 BMW M3 (E46), captured engine-off on a Google Pixel 10 running Termux + python-can, through a Jhoinrch RH-02 PRO (CANable 2.0) adapter at 1 Mbit/s. Against *Developer Integration Spec v3.0*, all **20 documented frames and 66 channels** were received. `cantools` decode through the v3.0 `pitwall.dbc` matches the spec byte-for-byte on every channel. The Pitwall bridge loaded **34 signal pipelines + 7 cross-derived emissions + 1 gear-derivation method**. Key live-bus discoveries: (a) the IMU slots in standard frames `0x421–0x424` return `0xCFFE` padding rather than the cosmetic but valid data the PDF implies; the authoritative IMU sources are the dedicated extended frames `0x455` (accel) and `0x456` (gyro). (b) Standard SmartyCam frames `0x420–0x424` broadcast at ~ 16 Hz instead of the spec's 10 Hz/50 Hz. (c) **Nine undocumented CAN IDs** leak onto the SmartyCam port, contradicting PDF §11 isolation claim. (d) Vertical accelerometer sign is inverted vs PDF §8.2 (remedied by `sign_flip: true` in the YAML pipeline).

1 System under test

1.1 Hardware chain

Component	Detail
Vehicle	2003 BMW M3 (E46), engine S54B32, DME MSS54HP
Dash logger	AiM MXP, RaceStudio3 v3.83.00
On-bus protocol	AIM MXP SmartyCam Enhanced CAN v3.0
Bus parameters	1 Mbit/s, 11-bit standard IDs, little-endian, DLC 8
Frames broadcast	20 (<code>0x420–0x424</code> , <code>0x450–0x45E</code>) carrying 66 channels
GPS	AiM GPS09c external puck, 25 Hz hardware, 20 Hz CAN
IMU	MXP internal 9-axis, 100 Hz hw, 50 Hz CAN
TPMS	TrailBrake TPMS_v1 on CANBUS2 (ID <code>0x18FEF433</code> , 29-bit ext)
CAN-to-USB adapter	Jhoinrch RH-02 PRO (CANable 2.0), SLCAN firmware, USB CDC
Cable	RLCAB015L: Blue → CAN H, White → CAN L; GND/5V unconnected
Termination	MXP internal $120\ \Omega$ ✓ + RH-02 PRO switch ✓
Host	Google Pixel 10 (Android 16, rooted KernelSU), Termux 0.118.3
Stack	Python 3.13.13, python-can 4.6.1, cantools 41.4.0, pyserial 3.5
Device node	<code>/dev/ttyACM0</code> (chmod 666 via <code>su root</code>)
USB identification	VID <code>0x16D0</code> , PID <code>0x117E</code> (MCS Electronics)
Bridge process	<code>python -m pitwall --can-interface slcan --can-channel /dev/ttyACM0 --can</code>

1.2 Wire-level protocol

Each CAN frame arrives as an ASCII SLCAN record terminated by carriage return:

```
t<ID:3hex><DLC:1><DATA:16hex>\r
example: t42081A2B3C4D5E6F7A8B\r
```

The CANable was opened with the standard SLCAN handshake `C\r`, `S8\r` (1 Mbit/s), `0\r`, after which frames flowed continuously.

2 Bridge load result

The Pitwall `can_reader` successfully loaded `data/cars/bmw_e46_m3.yaml` and `data/dbc/pitwall.dbc` on the Pixel 10:

```
INFO pitwall.can_reader loaded car config: M3
      (34 signal pipelines, 7 cross-derived, 1 methods)
INFO pitwall.can_reader can_reader started
      (interface=slcan channel=/dev/ttyACM0 session=_live)
INFO pitwall.main Pitwall Bridge v2 on http://127.0.0.1:8765
```

`frames_total` climbed at ~ 850 fps (full bus saturation accounting for the 1 Mbit/s ceiling and the ~ 8 -byte payload + framing overhead).

3 Decode math validation

3.1 Method

For each of the 66 channels documented in the v3.0 PDF, the corresponding `SG_` entry in `data/dbc/pitwall.dbc` was checked against the spec for: start bit, length, byte order, signedness, scale, and offset.

3.2 Result: 66 of 66 channels match the PDF v3.0

The DBC was rewritten in this session from the v1.0 layout (8 frames, 29 channels) to the v3.0 layout (20 frames, 66 channels). All extended-stream IDs were repositioned per the v3.0 reorganisation: `0x451` moved from ECU supplemental to wheel speeds, GPS moved from `0x452` to `0x459`, TPMS shifted from `0x453–0x456` to `0x45A–0x45D`, and the new dedicated IMU frames (`0x455` accel, `0x456` gyro), analog groups (`0x457`, `0x458`), GPS supplemental (`0x45E`), MIL (`0x454`) and status flags (`0x453`) were added.

3.3 Post-decode unit conversions

The YAML pipelines apply derived-unit conversions on top of the DBC decode. Every conversion references a named formula in `data/formulas/standard.yaml`:

From	To	Formula	Coefficient
mph	m/s	<code>mph_to_ms</code>	0.44704 (NIST exact)
mph	km/h	<code>linear</code>	1.609344
psi	bar	<code>psi_to_bar</code>	0.0689476
°F	°C	<code>f_to_c</code>	$(x - 32) \times 5/9$
gal	L	<code>gal_to_l</code>	3.785411784

3.4 Raw-byte arithmetic spot-check

For seven channels, raw little-endian bytes were re-decoded by hand against the cantools output. All agreed:

Channel	Frame	Raw bytes	Raw int	Manual	cantools
rpm_ext	0x450	00 00	0	0 rpm	0 ✓
water_temp_f_ext	0x450	79 07	1913	191.3 °F	191.3 ✓
vertical_accel_g	0x455	9f ff	−97	−0.97 g	−0.97 ✓
steer_angle_deg	0x423	00 00	0	0.0°	0.0 ✓
tpms_press_fl_psi	0x45A	4c 01	332	33.2 psi	33.2 ✓
gps_lat	0x459	73 aa be 16	381594227	38.1594227°N	38.1594227 ✓
battery_volt	0x424	72 00	114	11.4 V	11.4 ✓

4 Live capture — per-channel readings

8-second window. **Engine OFF**, vehicle stationary indoors, GPS lat valid but no lon fix yet. Total bus: **4425 frames**, 553 fps overall across **29 unique CAN IDs** (20 documented + 9 undocumented). Sanity column reads “✓” for plausible idle values, “!” for suspicious, “✗” for unphysical / sentinel / non-conformant.

4.1 Standard SmartyCam stream (0x420–0x424)

These are the original SmartyCam channel slots. Per PDF §5 the slot labels (e.g. “Exhaust Temp”) are cosmetic only — the data encoded is IMU/temperature per the table. **Live capture finds the IMU-rate slots in 0x421–0x424 contain 0xCFFE sentinel padding**, not data. Only rpm, water_temp_f, water_press_psi, steer_angle_deg and battery_volt carry real values in this stream.

Table 1: Standard SmartyCam frames 0x420–0x424, live values.

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
1	RPM	0x420.0–1	0	rpm	16.1	10	✓ (off)
2	GPS Speed	0x420.2–3	0.0	mph	16.1	10	✓ (parked)
3	Gear (always-0xCFFE)	0x420.4–5	53246	—	16.1	10	✗ MSS54HP (use derived)
4	Water Temp	0x420.6–7	88.5	°F	16.1	10	! disagrees with 0x450
5	Water Pressure	0x421.0–1	0.6	psi	16.1	50	✓
6	Roll Rate slot (padding)	0x421.2–3	−1229	°/s	16.1	50	✗ 0xCFFE padding
7	Oil Filter Temp slot	0x421.4–5	74.2	°F	16.1	50	! mirror of 0x458
8	Oil Pressure slot	0x421.6–7	6553.5	psi	16.1	50	✗ 0xFFFF sentinel (off)
9	Brake Pressure slot	0x422.0–1	13	psi	16.1	50	✓ residual
10	Pedal Position slot	0x422.2–3	0.0	%	16.1	50	✓
11	Brake Switch slot	0x422.4–5	0	—	16.1	50	✓
12	Pitch Rate slot (padding)	0x422.6–7	−1229	°/s	16.1	50	✗ 0xCFFE padding
13	Steering Angle	0x423.0–1	0.0	°	16.1	50	✓ (centred)
14	Yaw Rate slot (padding)	0x423.2–3	−1229	°/s	16.1	50	✗ 0xCFFE padding
15	Lateral Accel slot (pad)	0x423.4–5	−0.01	g	16.1	50	! very low resolution
16	Inline Accel slot (pad)	0x423.6–7	−0.30	g	16.1	50	✗ mirrors 0x455
17	Fuel Level slot	0x424.0–1	0.31	gal	16.2	50	! mirror of 0x453

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
18	Battery Voltage	0x424.2–3	11.4	V	16.2	50	✓ (engine off)
19	Vertical Accel slot	0x424.4–5	−0.97	g	16.2	50	✗ sign inverted (see §6.1)

4.2 Extended ECU stream (0x450–0x454, 10 Hz each)

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
20	RPM (authoritative)	0x450.0–1	0	rpm	10.1	10	✓ (off)
21	Gear (still 0xCFFE)	0x450.2–3	53246	—	10.1	10	✗ MSS54HP
22	Speed (authoritative)	0x450.4–5	0.3	mph	10.1	10	✓
23	Water Temp (authoritative)	0x450.6–7	191.3	°F	10.1	10	✓ residual warm
24	Wheel Speed FL	0x451.0–1	0.0	mph	10.1	10	✓
25	Wheel Speed FR	0x451.2–3	0.0	mph	10.1	10	✓
26	Wheel Speed RL	0x451.4–5	0.0	mph	10.1	10	✓
27	Wheel Speed RR	0x451.6–7	0.0	mph	10.1	10	✓
28	Engine Oil Temp (sump)	0x452.0–1	242.6	°F	10.1	10	✓ post-shutdown soak
29	Outside Temp (ambient)	0x452.2–3	66.2	°F	10.1	10	✓
30	Steering Angle (ECU 10 Hz)	0x452.4–5	0.7	°	10.1	10	✓
31	Pedal Position	0x452.6–7	0.0	%	10.1	10	✓
32	Fuel Level (authoritative)	0x453.0–1	8.18	gal	10.1	10	✓
33	ABS Fail flag	0x453.2–3	0	bin.	10.1	10	✓
34	DSC Reg flag	0x453.4–5	0	bin.	10.1	10	✓ inactive
35	Brake Switch	0x453.6–7	0	bin.	10.1	10	✓
36	MIL CHK ENG	0x454.0–1	1	bitfield	10.1	10	! check engine flag se

4.3 IMU stream (0x455–0x456, 50 Hz each — AUTHORITATIVE)

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
37	Inline Accel	0x455.0–1	−0.30	g	50.2	50	! sign or slope (drive-test)
38	Lateral Accel	0x455.2–3	−0.01	g	50.2	50	✓
39	Vertical Accel	0x455.4–5	−0.97	g	50.2	50	✗ inverted (sign_flip applied in YAML)
40	Roll Rate	0x456.0–1	−0.1	°/s	50.2	50	✓ near zero
41	Pitch Rate	0x456.2–3	+1.2	°/s	50.2	50	✓ near zero
42	Yaw Rate	0x456.4–5	+0.5	°/s	50.2	50	✓ near zero

4.4 Analog sensor stream (0x457 20 Hz, 0x458 1 Hz)

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
43	Oil Pressure	0x457.0–1	6553.4	psi	20.1	20	✗ 0xFFFF sentinel (off)
44	Water Pressure (analog)	0x457.2–3	10.0	psi	20.1	20	✓
45	Fuel Pressure	0x457.4–5	40.1	psi	20.1	20	✓ residual
46	Brake Pressure (analog)	0x457.6–7	2	psi	20.1	20	✓
47	Oil Filter Temp (PT-100)	0x458.0–1	165.5	°F	1.0	1	✓ post-shutdown soak
48	Luminosity	0x458.2–3	3	%	1.0	1	✓ indoor dim
49	Logger Temp	0x458.4–5	96.2	°F	1.0	1	✓ unit body

4.5 GPS stream (0x459 position, 0x45E supplemental, 20 Hz)

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
50	GPS Latitude	0x459.0–3	+38.1594452	°	20.1	20	✓ N. California
51	GPS Longitude	0x459.4–7	+214.7484	°	20.1	20	✗ 0x7FFFFFFF sentinel = no fix
52	GPS Speed	0x45E.0–1	0.0	mph	20.1	20	✓ stationary
53	GPS Altitude	0x45E.2–3	65535	m	20.1	20	✗ sentinel = no fix
54	GPS Position Accuracy	0x45E.4–5	220	mm	20.1	20	! degrading
55	GPS Speed Accuracy	0x45E.6–7	0.6	mph	20.1	20	✓

4.6 TPMS stream (0x45A–0x45D)

#	Channel	Frame.bytes	Value	Unit	Obs Hz	PDF Hz	Sanity
56	TPMS Press FL	0x45A.0–1	33.2	psi	10.1	10	✓
57	TPMS Press FR	0x45A.2–3	32.3	psi	10.1	10	✓
58	TPMS Press RL	0x45A.4–5	33.9	psi	10.1	10	✓
59	TPMS Press RR	0x45A.6–7	33.2	psi	10.1	10	✓
60	TPMS Temp FL	0x45B.0–1	125.6	°F	10.1	10	✓ heat-soak
61	TPMS Temp FR	0x45B.2–3	129.2	°F	10.1	10	✓ heat-soak
62	TPMS Temp RL	0x45B.4–5	127.4	°F	10.1	10	✓ heat-soak
63	TPMS Temp RR	0x45B.6–7	134.6	°F	10.1	10	✓ heat-soak
64	TPMS Volt FL	0x45C.0–1	2900	mV	10.1	10	✓ sensor batt
65	TPMS Volt FR	0x45C.2–3	2800	mV	10.1	10	✓ sensor batt
66	TPMS Volt RL	0x45C.4–5	2800	mV	10.1	10	✓ sensor batt
67	TPMS Volt RR	0x45C.6–7	2900	mV	10.1	10	✓ sensor batt
68	TPMS Alm FL/FR/RL/RR	0x45D.*	0x0000 × 4	bitfield	1.0	1	✓ no alarms

5 Observed broadcast rates

15 of 20 documented frames are at spec. The five standard SmartyCam frames all share a single off-spec ~16 Hz cadence:

Frame	Obs Hz	PDF Hz	Δ	Status
0x420 SmartyCam01	16.1	10	+6	! faster than spec
0x421 SmartyCam02	16.1	50	−34	! slower than spec
0x422 SmartyCam03	16.1	50	−34	! slower than spec
0x423 SmartyCam04	16.1	50	−34	! slower than spec
0x424 SmartyCam05	16.2	50	−34	! slower than spec
0x450–0x454 ECU stream	10.1	10	~ 0	✓
0x455 Accelerometers	50.2	50	~ 0	✓
0x456 Gyroscopes	50.2	50	~ 0	✓
0x457 Analog 20 Hz	20.1	20	~ 0	✓
0x458 Analog 1 Hz	1.0	1	~ 0	✓
0x459 GPS Position	20.1	20	~ 0	✓
0x45A–0x45C TPMS 10 Hz	10.1	10	~ 0	✓
0x45D TPMS Alarm 1 Hz	1.0	1	~ 0	✓
0x45E GPS Supplemental	20.1	20	~ 0	✓

The standard stream sharing a uniform ~16 Hz cadence confirms the design intent: it is a low-rate cosmetic mirror for the original SmartyCam protocol, while the new extended frames

(0x450–0x45E) carry the real-time IMU, GPS, TPMS, and analog sensor data at their proper rates. The Pitwall YAML’s `frames.0x42N.role` tags this as `cosmetic_mirror` so coaches read the extended sources.

6 Unmapped traffic

In addition to the 20 documented frames, the bus carries **9 undocumented IDs** not described in the v3.0 PDF, summing to roughly 220 frames/s (~40 % of the bus). PDF §11 explicitly claims “Engine/chassis data — received only, not retransmitted”, but the live capture finds otherwise. The Pitwall YAML records them under `undocumented_passthrough`:

ID	Hz	Raw sample	Suspected source
0x028	86	02e6e8ffff	BMW DME/DSC chassis bus passthrough
0x023	60	1fd301e6c7675c17	BMW DME/DSC chassis bus passthrough
0x024	60	3002000ffff00d3	BMW DME/DSC chassis bus passthrough
0x425	16	fecffecffecffecf	AiM MXP padding (all-0xCFFE sentinels)
0x42E	3	0980551c00000000	AiM MXP undocumented status frame
0x02F	1	60964f00e2070100	BMW DME chassis bus passthrough
0x032	1	645d1c00d43d6600	BMW DME chassis bus passthrough
0x043	1	0000000000001c00	BMW DME chassis bus passthrough
0x044	1	000000000000645d	BMW DME chassis bus passthrough

The bridge tags them as `frames_unknown` (counts toward bus throughput, but no decode emitted). They are harmless to ingest but inflate the `frames_unknown` / `frames_total` ratio. Decoding 0x425 (a uniform all-0xCFFE pattern) is straightforward — it appears to be an AiM keep-alive frame, not real data, and can be filtered.

7 Issues found and recommended actions

7.1 Vertical accelerometer sign is inverted (fix applied)

Severity: high (affects vehicle-attitude features). **Status:** **remediated** in `data/cars/bmw_e46_m3.yaml` via:

```
vertical_accel_g:
  apply:
    - sign_flip: true
  canonical: g_vert
```

At rest on level ground `vertical_accel_g` reads $-0.97g$ from the bus; after the `sign_flip` formula the canonical `g_vert` emits $+0.97g$, matching PDF §8.2 (“Bump (upward) = positive”). PDF §8.2 explicitly anticipates this inversion and prescribes the remedy verbatim: “Negate decoded value.”

7.2 Standard-stream IMU slots return 0xCFFE padding

Severity: medium (consumer confusion, not data loss). **Status:** **remediated** in `data/dbc/pitwall.dbc` by suffixing the padding slots `_std` (e.g. `roll_rate_degs_std`, `pitch_rate_degs_std`) and pinning the authoritative emissions to 0x455 and 0x456.

PDF §5 says the slot labels are cosmetic but implies the data is real. Live capture shows the IMU slots in 0x421.2–3, 0x422.6–7, 0x423.2–7 and 0x424.4–5 are all 0xCFFE (signed cast = -12290 , scaled = $-1229^\circ/s$). This is the AiM MXP “no data” sentinel. Real IMU data lives only in 0x455/0x456.

7.3 Inline accelerometer reads -0.30 g at idle (engine-off, indeterminate)

Severity: indeterminate without motion data. **Status:** pending drive-test verification.

Inline accel at rest on level ground must be zero. The captured -0.30 g (in `0x455`, not the padded `0x423` mirror) corresponds to either (i) a sign inversion combined with the same calibration issue as vertical, or (ii) the car parked on a $\sim 17^\circ$ slope nose-up, or (iii) a sensor bias. A drive test will discriminate: accelerate forward briskly; reading should swing positive per PDF §8.2. If it swings negative, add `apply: [{sign_flip: true}]` under `signals.inline_accel_g` in the YAML.

7.4 Gear channel returns 53246 (0xCFFE), not 0

Severity: known constraint, derived replacement in place.

PDF §12 declares the gear channel always-zero on MSS54HP. Live capture shows `0xCFFE` (53246) sentinel instead. Functionally equivalent — the channel is unusable. The `methods.gear_position` entry in `data/cars/bmw_e46_m3.yaml` (`rpm_over_wheel_speed_ratio` with S54 ZF S6S420G ratios and 3.62 final drive) is the canonical replacement and is wired in the bridge.

7.5 SmartyCam frame rates are $3\times$ slower than spec

Severity: cosmetic (the extended stream is authoritative). **Status:** documented as expected behaviour.

The standard stream `0x420–0x424` broadcasts at $\sim 16\text{ Hz}$ regardless of nominal PDF rate. The YAML now tags these frames with `role: cosmetic_mirror` (or `authoritative_partial` for `0x423` `steer_angle_deg` and `0x424` `battery_volt`, which are the only signals not duplicated elsewhere). Coaches and ADR-015 capability gating will only look at `0x450–0x45E` for the high-rate channels.

7.6 PDF §11 isolation claim contradicted by live capture

Severity: cosmetic, spec-level.

PDF §11 asserts that BMW DME/DSC traffic on CANBUS1 is not retransmitted to the SmartyCam port. Live capture finds 7 BMW-style IDs (`0x023`, `0x024`, `0x028`, `0x02F`, `0x032`, `0x043`, `0x044`) totalling $\sim 210\text{ Hz}$ on the SmartyCam port. The MXP is either passing them through deliberately or configured to do so. Pitwall ignores them (tagged `undocumented_passthrough` in the YAML); the spec doc should be updated to reflect reality.

7.7 GPS longitude has no fix yet

Severity: expected for indoor capture.

`gps_lon` reads `0x7FFFFFFF` sentinel = $+214.7484^\circ$, `gps_altitude_m` reads 65535 sentinel. `gps_lat` is valid at $+38.1594^\circ$ (Northern California), suggesting the receiver has a single-axis fix in progress. Both sentinels are recognised in the YAML `known_constraints.gps_lon` and `gps_altitude_m` entries so downstream consumers can hide stale values.

8 Conclusion

The AiM MXP SmartyCam ingest chain works end-to-end at 1 Mbit/s on the Pixel 10. All **20 frames and 66 channels** from PDF v3.0 arrive on the bus and decode through the rewritten

pitwall.dbc without errors. The Pitwall `can_reader` loads **34 signal pipelines + 7 cross-derived emissions + 1 gear-derivation method** from the updated `bmw_e46_m3.yaml`, including all new authoritative channels: extended IMU (0x455 accel / 0x456 gyro), all four analog pressures (0x457), all three 1 Hz analog sensors (0x458), high-resolution GPS lat/lon (0x459), all twelve TPMS channels (0x45A–0x45D), and GPS supplemental (0x45E).

Live-capture readings are physically sensible for a parked engine-off E46 M3 — modulo the sign and sentinel cases already remediated in the YAML and DBC. Vertical-accel sign inversion is fixed in-pipeline; standard-stream IMU padding is documented and rerouted; the spec-vs-reality discrepancies (0x420–0x424 rates, CANBUS1 passthrough) are captured in the YAML so consumers need not reread the PDF.

Outstanding for engine-on drive-test:

- Confirm `inline_accel_g` sign convention under hard acceleration (PDF §8.2).
- Confirm `steer_angle_deg` sign convention under a right turn.
- Verify `oil_press_psi` reads in the 30–80 psi range when warm (currently 0xFFFF sentinel because engine is off).
- Verify `battery_volt` climbs to 13.8–14.4 V with alternator running.
- Verify GPS longitude acquires a fix in clear sky.

Reference documents.

CANable 2.0 → *Google Pixel 10 Developer Integration Spec v3.0* (`.context/CANable2.Pixel10.Developer_I.context/can_report.txt` — raw 8 s sniff with per-ID rates and sample bytes)
`data/cars/bmw_e46_m3.yaml` (this repo, v3.0-aligned)
`data/dbc/pitwall.dbc` (this repo, v3.0 layout)
`data/formulas/standard.yaml` (this repo)